

Matrix Keypad

Scan a physical matrix; interpret one deliberate key

D13 and TP-R0 explain the circuit without a Serial monitor.

Lesson	016 — component	Time	120–180 minutes
Prerequisites	Lessons 001–015	Board	Arduino Mega 2560
Evidence	D13 patterns and TP-R0	Status	host verified; bench open
Safety	E1: USB-powered passive input circuit only		

1 Mission

Build one operator-input path:

key matrix → `MatrixKeypad` → `KeypadSnapshot` → D13 evidence.

`MatrixKeypad` owns and scans seven pins. Its composed `Keypad` interpreter debounces the 12-bit sample, maps one position to a semantic key, and rejects chords until every key is released. Time enters only through `update(now)`.

Boundary. This lesson teaches input interpretation. It is not a lock, credential system, or anti-ghosting keypad. A diode-less matrix can produce ambiguous multi-key observations, so every chord is rejected.

1.1 Learning outcomes

- explain row selection and column pull-ups;
- separate electrical scanning from deterministic interpretation;
- predict debounce, press, release, and invalid-chord events;
- distinguish acquisition evidence from safe-state evidence;
- use RAII cleanup without exceptions, heap allocation, or blocking delay.

2 Parts and safety

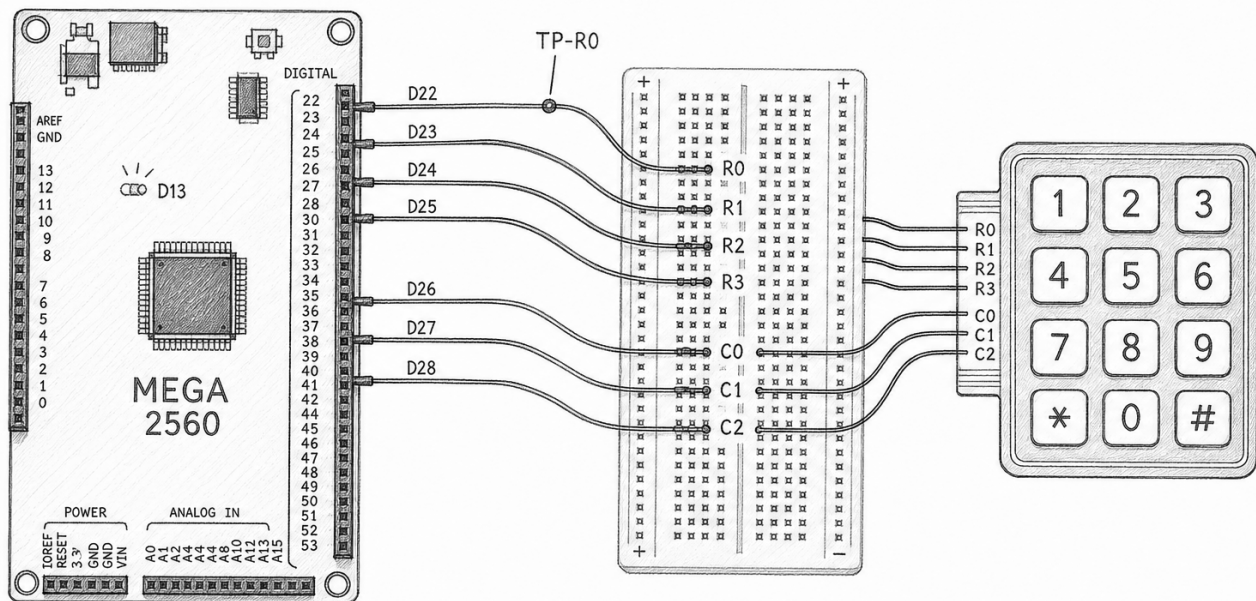
Part	Requirement
Arduino Mega 2560	USB-powered board
4x3 membrane keypad	Passive switch matrix with identified seven-wire tail
Seven jumpers	D22 through D28
Logic probe or oscilloscope	Optional TP-R0 observation; high-impedance input

Remove USB power before moving wires or attaching a probe. Identify the exact tail order from manufacturer documentation or an unpowered continuity test; printed connector order is not standardized. No keypad conductor connects directly to 5 V or GND.

3 Authoritative wiring

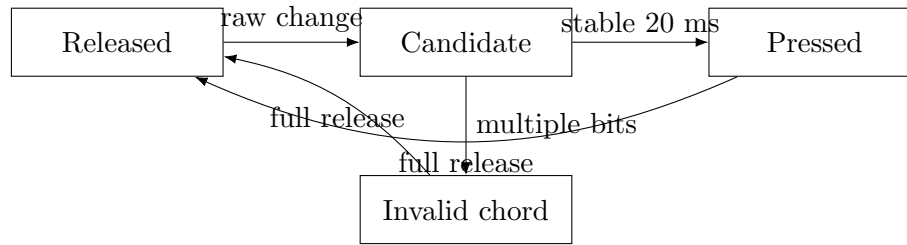
Signal	Mega pin	Keypad contact
Row0 / TP-R0	D22	row containing 1, 2, 3
Row1	D23	row containing 4, 5, 6
Row2	D24	row containing 7, 8, 9
Row3	D25	row containing *, 0, #
Column0	D26	column containing 1, 4, 7, *
Column1	D27	column containing 2, 5, 8, 0
Column2	D28	column containing 3, 6, 9, #
Operator evidence	D13	board LED; no external wire

During each scan, exactly one row becomes an output driven low. Columns remain inputs with internal pull-ups. A closed key pulls its column low. The selected row immediately returns to input before the next row is selected.



Alternative text: pencil orientation plate with a Mega on the left, a breadboard in the middle, and a 4x3 keypad on the right; seven wires connect D22–D28 to rows R0–R3 and columns C0–C2, with TP-R0 on D22. The table above, not the illustrative connector order, is authoritative.

4 Scan and interpretation



A stable single bit produces one `pressEvent`. Holding the key produces no repeats. A full stable release rearms the next press. A chord that collapses to one key remains disarmed until complete release.

Observation	D13 pattern	Interpretation
Initialization	one short pulse	keypad and LED claims succeeded
Digit 1–9	one to nine short pulses	accepted semantic digit
Digit 0	ten short pulses	accepted zero, distinct from ready
*	one long pulse	accepted star
#	two long pulses	accepted hash
Invalid chord	repeating two-short group	release every key before retry
Scan/update fault	repeating three-short group	stop relying on input

5 Narrative code

```

void loop ()
{
  if (!running)
  {
    return;
  }

  const adk::TimePoint      now      (millis ());
  const adk::KeypadSnapshot observation = observeKeypad      (now);
  const OperatorDecision    decision  = decideOperatorEvent (observation);

  if (!showOperatorEvidence (now, decision))
  {
    stopSafely ();
  }
}

```

The canonical sketch acquires the keypad, then the evidence LED. On a partial failure it releases the earlier object. The loop always reads “observe, decide, actuate.” D13 patterns are nonblocking, so scanning continues while a result is displayed. `stopSafely()` shuts down in reverse order.

6 Experiments

6.1 Experiment A: acquisition and row scan

Predict. Before successful initialization, TP-R0 has no recurring selection pulse. During scanning, D22 has one bounded low interval per update.

Observe. Attach the probe to D22 and GND while unpowered. Restore USB power and record period _____ and low time _____.

Interpret. A recurring bounded pulse supports the claim that scanning began after acquisition. Its absence does not identify which claim failed.

6.2 Experiment B: mapping and release gate

Predict the D13 pattern, press one key, hold it, then release it completely.

Key	Predicted pattern	Observed pattern	One event while held?
1	_____	_____	_____
0	_____	_____	_____
*	_____	_____	_____
#	_____	_____	_____

6.3 Experiment C: invalid chord

Predict. Two held keys produce the repeating paired-short indication. Releasing only one key does not fabricate a new accepted press.

Observe. Hold two keys, release one, then release the other. Result: _____.

Interpret. Acceptance only after a complete release demonstrates release gating; it does not prove arbitrary combinations are ghost-free.

6.4 Experiment D: safe state

After the shutdown exercise, predict that D22–D28 are inputs, TP-R0 pulses stop, column pull-ups are removed, and D13 is inactive. Observe TP-R0: _____. D13 off alone does not prove keypad claims were released.

7 Diagnosis

Observation	Check
No ready pulse and no TP-R0 pulse	pin conflict, initialization status, and D13 ownership
Ready pulse but no accepted key	tail order, continuity, and row/column mapping
Wrong digit	logical row/column order; do not rewrite the semantic map to hide wiring
Paired pattern for one key	unintended bridge or ambiguous tail identification
Three-short pattern	update status and lifecycle; do not treat the snapshot as input
TP-R0 continues after shutdown	safe-state failure; remove USB power and record it

8 CLI and acceptance

```
make host-test-keypad
make host-test-matrix-keypad
make arduino-Lesson016MatrixKeypad
make firmware-size EXAMPLE=Lesson016MatrixKeypad
make lessons
make lessons-check
```

Software record: commit _____; Arduino CLI _____; AVR core _____;
flash _____; static RAM _____.

Bench card—open until measured

- ☐ Exact Mega revision and keypad part recorded.
- ☐ Tail order identified while unpowered.
- ☐ D22–D28 inspected against the table.
- ☐ TP-R0 scan cadence measured.
- ☐ All twelve keys and invalid chord observed.
- ☐ Reset, shutdown, and USB power removal observed.
- ☐ D22–D28 safe state checked separately from D13.

Supply _____; instrument _____; date _____; reviewer _____.
Deviations: _____.

Current result: host verified; hardware acceptance open. Compilation, deterministic tests, and this worksheet do not claim physical verification. Record measured evidence before changing that status.